Constraints on Higgs-charm couplings at CMS

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on behalf of the CMS Collaboration

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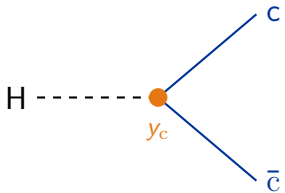
Higgs 2026, Louvain-la-Neuve, 28 October 2026

Why the Higgs-charm coupling?

- Higgs couplings to the 3rd generation (t, b, τ) and to μ are established
- **2nd-generation quark Yukawas are the open frontier**: charm is the most accessible
  - SM: $\mathcal{B}(\mathrm{H} \to \mathrm{c\bar{c}}) \approx 2.9\%$, about 20× below $\mathrm{H} \to \mathrm{b\bar{b}}$
  - Huge QCD backgrounds, c-jets hard to tell apart from b- and light jets
- BSM: flavour-compatible models allow $|\kappa_\mathrm{c}|$ of order 3

  e.g. spontaneous flavour violation, vector-like quarks (arXiv:2410.05236)

  $\Rightarrow$ current sensitivity is now entering this region
- Is the Higgs mechanism also responsible for the 2nd-generation quark masses?



Coupling modifier

$$\kappa_\mathrm{c} = y_\mathrm{c} / y_\mathrm{c}^\mathrm{SM}$$

(κ framework; SM: $\kappa_\mathrm{c} = 1$)

How CMS probes κ_c

Direct $H \to c\bar{c}$ (rate $\propto \kappa_c^2/\Gamma_H$)

VH($H \to c\bar{c}$)	PRL 131 (2023) 061801
$t\bar{t}H(H \to c\bar{c})$ + VH comb.	PRL 136 (2026) 011801
Boosted $H \to c\bar{c}$ (mostly ggH)	PRL 131 (2023) 041801

best: $|\kappa_c| < 3.5$ (2.7 exp.)

H + charm production (cg $\to$ cH)

cH($H \to \gamma\gamma$)	JHEP 11 (2025) 060
cH($H \to WW$) + comb.	JHEP 08 (2026) 051

best: $|\kappa_c| < 47$ (51 exp.)

Indirect: H production and width

γH + on-shell $H \to 4\ell$	PRD 112 (2025) 112001
Off-shell + on-shell $H \to 4\ell$ **NEW**	arXiv:2607.23352

$\kappa_c \in [-3.7, 3.2]$ to $[-8.6, 7.5]$ (by assumptions)

Rare exclusive decays

$H \to J/\psi\gamma$, $\psi(2S)\gamma$	PLB 865 (2025) 139462

sensitive to the *sign* of κ_c; details in J. Bierkens' talk

$-166 < \kappa_c < 208$

All results with 13 TeV Run 2 data, up to 138 fb^{-1}. **NEW** = public in 2026.

Direct $\mathrm{H} \to \mathrm{c\bar{c}}$: VH and boosted H

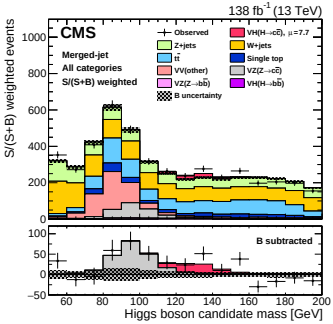
- **VH**: leptonic W/Z, resolved + merged-jet (ParticleNet) topologies
- VZ(Z→c$\bar{\text{c}}$) observed, 5.7σ

VH($\mathrm{H} \to \mathrm{c\bar{c}}$), 138 fb^{-1}

$\mu < 14\,(7.6)$
$1.1 < |\kappa_\mathrm{c}| < 5.5$ $\quad(|\kappa_\mathrm{c}| < 3.4$ exp.)

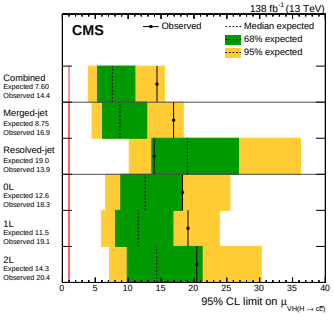
Boosted $\mathrm{H} \to \mathrm{c\bar{c}}$, $p_\mathrm{T} > 450$ GeV

$\mu < 47\,(39)$ (mostly ggH)



VH merged-jet: $m(\mathrm{H_{cand}})$, S/(S+B) weighted

PRL 131 (2023) 061801; PRL 131 (2023) 041801

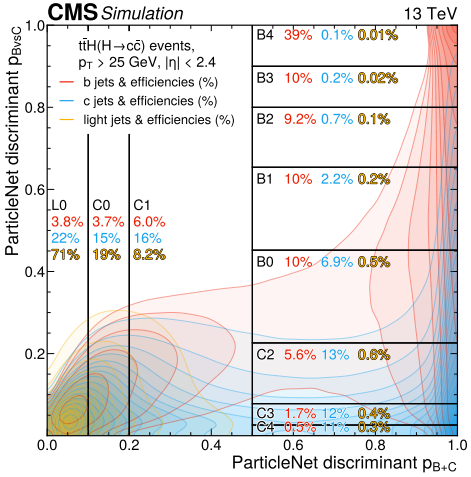


VH: μ limits per channel

$t\bar{t}H(H \to c\bar{c})$: a new channel

PRL 136 (2026) 011801, CMS-HIG-24-018

- First search for $H \to c\bar{c}$ in $t\bar{t}H$; 0, 1, 2 lepton $t\bar{t}$ decays
- Many jets of all flavours: ParticleNet b vs. c vs. light **2D categories**
- ParT event classifier separating $t\bar{t}H$(cc), $t\bar{t}H$(bb), $t\bar{t}Z$, $t\bar{t}$+HF
- **Simultaneous** fit of $t\bar{t}H(H \to c\bar{c})$, $t\bar{t}H(H \to b\bar{b})$, $t\bar{t}Z(Z \to b\bar{b}/c\bar{c})$
  - $t\bar{t}$+cc/bb normalised in dedicated regions
  - $t\bar{t}Z(Z \to c\bar{c})$ as a standard candle: $\mu = 1.0 \pm 0.8$
- By-product: $t\bar{t}H(H \to b\bar{b})$ $\mu = 0.91^{+0.26}_{-0.22}$, 4.4σ (4.5 exp.)



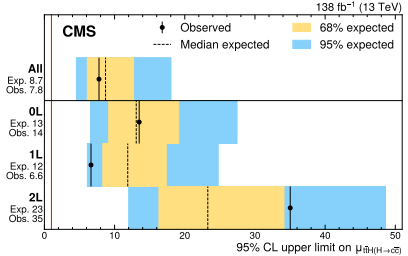
b, c, light jets in the ParticleNet (b vs. light, c vs. light) plane and the 2D tagging categories

$\mathrm{tt}\mathrm{H}(\mathrm{H} \to \mathrm{c}\bar{\mathrm{c}})$: results

$\mathrm{tt}\mathrm{H}(\mathrm{H} \to \mathrm{c}\bar{\mathrm{c}})$, 138 fb^{-1}

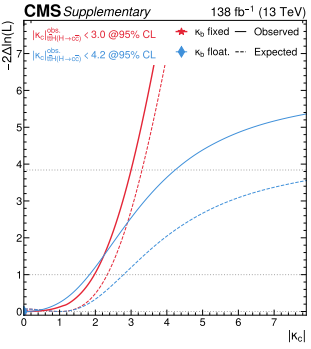
$\mu_{\mathrm{ttH(cc)}} < 7.8\,(8.7)$
$\sigma\mathcal{B} < 0.11\,(0.13)$ pb
$|\kappa_\mathrm{c}| < 3.0\,(3.3)$, $\kappa_\mathrm{b} = 1$
$|\kappa_\mathrm{c}| < 4.2$ obs., κ_b floating

- Expected sensitivity comparable to VH($\mathrm{H} \to \mathrm{c}\bar{\mathrm{c}}$)
- Simultaneous bb/cc fit constrains κ_b and κ_c together



$\mu_{\mathrm{ttH(cc)}}$ limits per channel

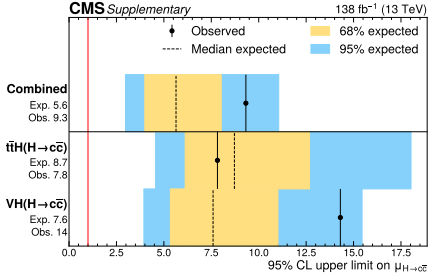
PRL 136 (2026) 011801, CMS-HIG-24-018

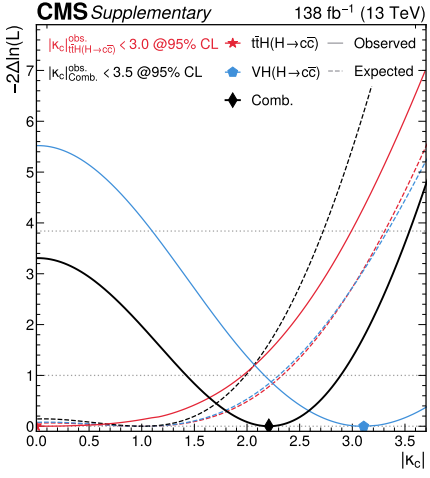


$|\kappa_\mathrm{c}|$ scan, κ_b fixed / floating

Combination of ttH and VH: tightest constraint on κ_c

PRL 136 (2026) 011801, CMS-HIG-24-018





ttH + VH

$\mu_{H\to c\bar{c}} < 9.3$
(5.6 exp.)

$|\kappa_c| < 3.5$
(2.7 exp.)

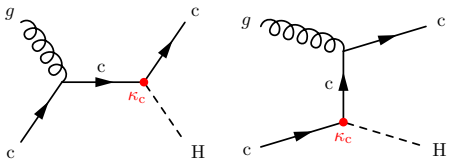
Expected limit improves by ∼20% over the best single channel

Most stringent direct constraint on κ_c to date; expected reach now below the BSM-motivated $|\kappa_c| \sim 3$

H + charm production: κ_c in production

- κ_c enters the **production** vertex: independent of $H \to c\bar{c}$-decay; clean H decays + a c-tagged jet
- $\sigma_{cH} = 90\,\text{fb}$: $\sigma\mathcal{B} \approx 0.2\,\text{fb}$ ($\gamma\gamma$), $\approx 20\,\text{fb}$ (WW)
- Challenges
  - H+c also arises from g→c$\bar{c}$ splitting in ggH: κ_c-**independent**, nearly identical final state
  - soft, often forward c-jets
  - flavour-scheme modelling of cH (→ T. Bevilacqua, next talk)
- κ_c from "flat direction": other H rates kept SM-like

JHEP 11 (2025) 060; JHEP 08 (2026) 051



LO diagrams for cH production; κ_c enters at the marked vertices

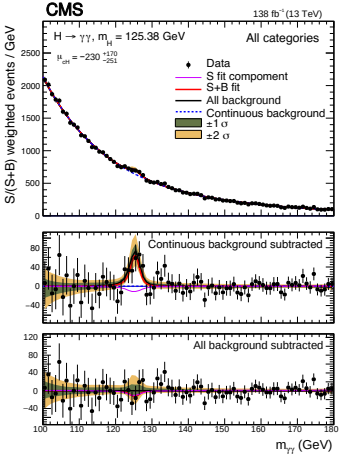
cH($H\to\gamma\gamma$)

JHEP 11 (2025) 060, CMS-HIG-23-010

- First search for cH with $H\to\gamma\gamma$, 138 fb^{-1}
- Two BDTs, trained without c-tag inputs:
  - BDT1: cH vs. ggH (largest resonant background)
  - BDT2: cH vs. continuum $\gamma\gamma$+jets
- 9 categories per year (27 in total); signal from fits to $m_{\gamma\gamma}$

Result, 95% CL

$\mu_{\mathrm{cH}} < 243\,(355)$
$|\kappa_{\mathrm{c}}| < 38.1\,(72.5)$



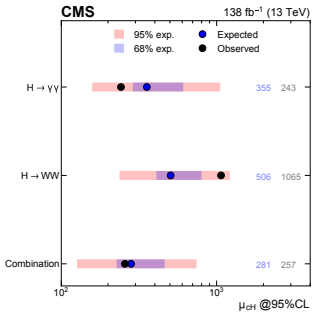
$m_{\gamma\gamma}$, all categories, S/(S+B) weighted

cH(H→WW) and combination

- H→WW→e$\nu\mu\nu$ + c-jet; SRs with 1 or >1 c-jet, CRs for $t\bar{t}$ and high $m_{\ell\ell}$
- Two BDTs: vs. non-H backgrounds and vs. other H processes
- H+c background (κ_c-independent) split by jet flavour; correlation with cH: −87%
- 1-POI fit: $\mu_{\mathrm{cH}} < 1065\,(506)$, $|\kappa_c| < 211\,(95)$
  2-POI (H+c bkg floating): $< 1804\,(1097)$

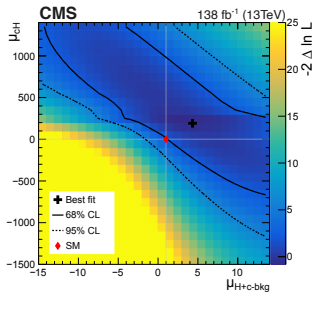
$\gamma\gamma$ + WW combination

$\mu_{\mathrm{cH}} < 257\,(281)$
$|\kappa_c| < 47\,(51)$



μ_{cH} limits: $\gamma\gamma$, WW, combined

JHEP 08 (2026) 051, CMS-HIG-24-009

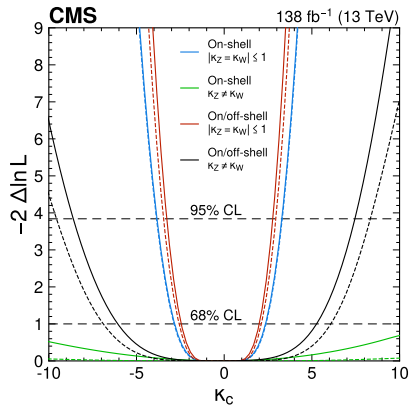


WW: μ_{cH} vs. $\mu_{\mathrm{H+c\ bkg}}$

Indirect constraints on κ_c

PRD 112 (2025) 112001; arXiv:2607.23352; PLB 865 (2025) 139462

- Large light-quark Yukawas enhance $q\bar{q}\to H$ and γH production and change Γ_H
- γ**H + on-shell H**$\to 4\ell$: $\kappa_c \in [-3.7, 3.2]$ ($[-3.8, 3.3]$)
  other light Yukawas floating, $|\kappa_V| \leq 1$
- NEW **Off-shell + on-shell H**$\to 4\ell$: off-shell rate breaks the width degeneracy
  - no assumption on κ_Z, κ_W: $\kappa_c \in [-8.6, 7.5]$ ($[-9.6, 8.3]$)
  - with $|\kappa_V| \leq 1$: $\kappa_c \in [-3.3, 2.8]$ ($[-3.5, 2.9]$)
  - width measurement: see F. Errico's talk
- **H**$\to$**J/**$\psi\gamma$: $-166 < \kappa_c < 208$ (see J. Bierkens' talk)



arXiv:2607.23352: κ_C scans, on-shell and on+off-shell, with and without $|\kappa_V| \leq 1$

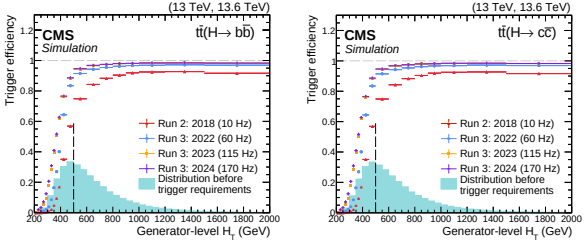
Where we stand: CMS constraints on κ_c

Probe	Reference	κ_c at 95% CL: obs. (exp.)	Main assumptions
$t\bar{t}H + VH(H \to c\bar{c})$	PRL 136 (2026) 011801	$\lvert\kappa_c\rvert < 3.5$ (2.7)	$\kappa_b = 1$, SM-like width
$t\bar{t}H(H \to c\bar{c})$	PRL 136 (2026) 011801	$\lvert\kappa_c\rvert < 3.0$ (3.3)	$\kappa_b = 1$
$VH(H \to c\bar{c})$	PRL 131 (2023) 061801	$1.1 < \lvert\kappa_c\rvert < 5.5$ (3.4)	$\kappa_b = 1$
$\gamma H + H \to 4\ell$	PRD 112 (2025) 112001	$[-3.7, 3.2]$ ($[-3.8, 3.3]$)	$\lvert\kappa_V\rvert \leq 1$
On+off-shell $H \to 4\ell$	arXiv:2607.23352	$[-8.6, 7.5]$ ($[-9.6, 8.3]$)	none on $\kappa_{Z,W}$
cH ($\gamma\gamma$ + WW)	JHEP 08 (2026) 051	$\lvert\kappa_c\rvert < 47$ (51)	
$H \to J/\psi\gamma$	PLB 865 (2025) 139462	$[-166, 208]$	$\kappa_\gamma = 1$

- Direct $H \to c\bar{c}$ searches lead; indirect probes add sensitivity to the **sign** of κ_c and different model assumptions
- Several probes now reach $|\kappa_c| \sim 3$, where flavour-compatible BSM scenarios live

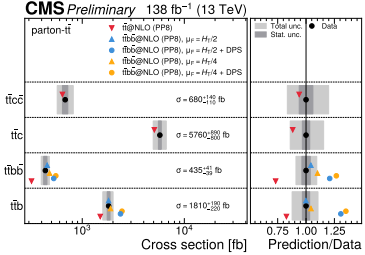
Toward Run 3: tools already in place

Heavy-flavour triggers



arXiv:2608.18331 (CMS-BTV-25-002): ML b/c tagging at the HLT; $\mathrm{tt}\mathrm{H}(\mathrm{H} \to \mathrm{c}\bar{\mathrm{c}})$ trigger efficiency, Run 2 vs. 2022–24

$\mathrm{tt}+\mathrm{c}\bar{\mathrm{c}}$ background measured



CMS-PAS-TOP-26-003 **NEW**:

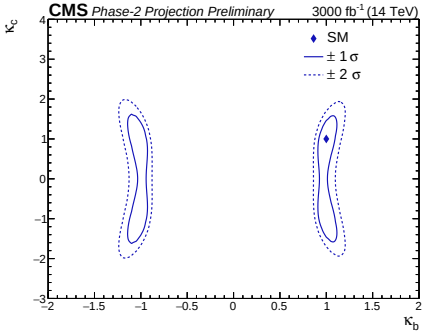
$$\sigma_{\mathrm{tt}\bar{\mathrm{c}}\mathrm{c}} = 680^{+130}_{-100}\,(\mathrm{syst}) \pm 40\,(\mathrm{stat})\,\mathrm{fb}$$

- Run 3 at 13.6 TeV: dataset already larger than Run 2, new taggers (UParT, GloParT), parking and scouting streams for hadronic final states
- Run 3 analyses are under way in all channels

Prospects at the HL-LHC

- ATLAS + CMS input to the European Strategy update
  (arXiv:2504.00672, CMS-HIG-25-002), 3 ab^{-1} per experiment:
  - VH($\mathrm{H} \to \mathrm{c\bar{c}}$): $\approx 1.6\sigma$
  - $|\kappa_\mathrm{c}| < 1.5$ at 95% CL, 30% better than previous projections
- History: expected $\mu_\mathrm{H} \to \mathrm{c\bar{c}}$ limit improved from $\sim$37 (2016 data) to 5.6 (Run 2 VH+$\mathrm{t\bar{t}H}$)
  far faster than luminosity alone: taggers, new channels, analysis techniques
- Further gains: Run 3 data, timing and track-trigger upgrades, ML tagging; theory modelling and tagger calibration become limiting

Is $\mathrm{H} \to \mathrm{c\bar{c}}$ evidence reachable at the HL-LHC?
Determining $|\kappa_\mathrm{c}|$ at $\mathcal{O}(1)$ is.



CMS-HIG-21-008 (supplementary): expected (κ_b, κ_c) contours at 3000 fb^{-1}, 14 TeV, extrapolated from the Run 2 VH($\mathrm{H} \to \mathrm{c\bar{c}}$) analysis

Summary

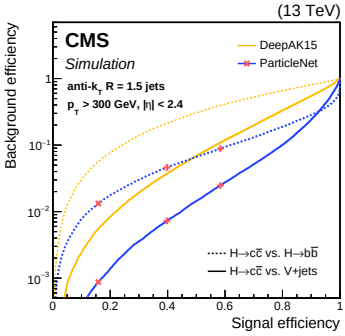
- The charm Yukawa is the most accessible 2nd-generation quark coupling, and CMS probes it in many complementary ways
- Direct searches: $\mathrm{t\bar{t}H(H \to c\bar{c})}$ combined with $\mathrm{VH(H \to c\bar{c})}$
  $\Rightarrow |\kappa_c| < 3.5$ **(2.7 exp.)**, the most stringent constraint to date
- H+c production probes κ_c independently of $\mathrm{H \to c\bar{c}}$: $\gamma\gamma$ + WW give $|\kappa_c| < 47$ (51 exp.)
- Indirect probes (γH, on/off-shell H$\to 4\ell$, H$\to$J/$\psi\gamma$) reach $|\kappa_c| \sim 3$ under different assumptions; off-shell H$\to 4\ell$ removes the assumption on $\kappa_{Z,W}$
- Sensitivity is now reaching the BSM-motivated region $|\kappa_c| \sim 3$
- Run 3: new HLT tagging and a measured $\mathrm{t\bar{t}+c\bar{c}}$ background; HL-LHC projects $|\kappa_c| < 1.5$

Thank you!

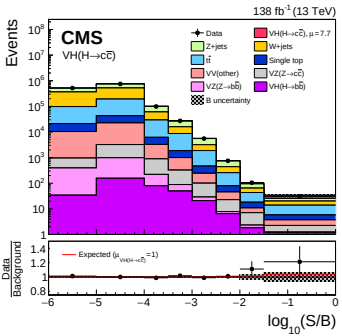
Backup

VH($\mathrm{H} \to \mathrm{c\bar{c}}$): event categories and sensitivity

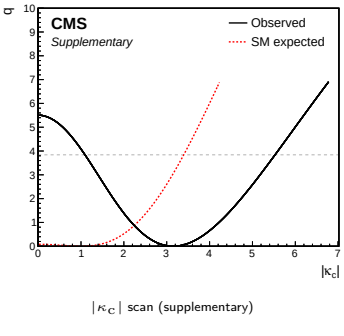
PRL 131 (2023) 061801



ParticleNet merged-jet $\mathrm{H} \to \mathrm{c\bar{c}}$-tagging



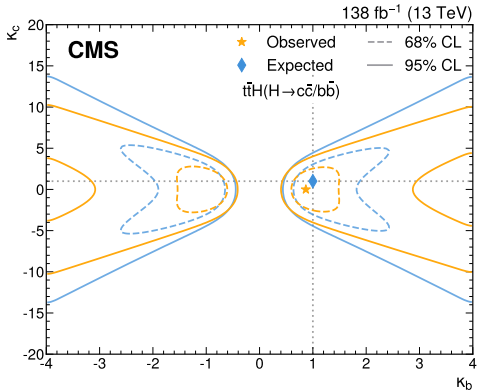
Events vs. $\log_{10}(S/B)$



$|\kappa_c|$ scan (supplementary)

Best fit $\mu = 7.7^{+3.8}_{-3.5}$; merged $\mu = 8.7^{+4.6}_{-4.0}$, resolved $\mu = -9.5 \pm 9.6$. $\sigma\mathcal{B} < 0.94$ (0.50) pb.

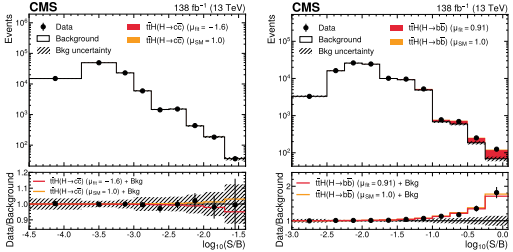
$\mathrm{t\bar{t}H(H \to c\bar{c}/H \to b\bar{b})}$: (κ_b, κ_c) plane



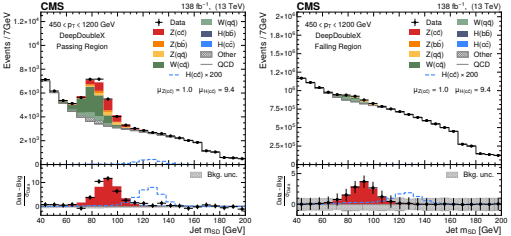
PRL 136 (2026) 011801

- Simultaneous $\mathrm{t\bar{t}H(H \to b\bar{b})}$ + $\mathrm{t\bar{t}H(H \to c\bar{c})}$ fit constrains both Yukawas
- Signal strengths: $\mathrm{t\bar{t}H}$(bb) $0.91^{+0.26}_{-0.22}$; $\mathrm{t\bar{t}H}$(cc) -1.6 ± 4.5; $\mathrm{t\bar{t}Z}$(bb) $1.47^{+0.45}_{-0.41}$; $\mathrm{t\bar{t}Z}$(cc) $1.02^{+0.79}_{-0.84}$

$\mathrm{t\bar{t}H(H \to c\bar{c})}$ and boosted $\mathrm{H \to c\bar{c}}$: more figures

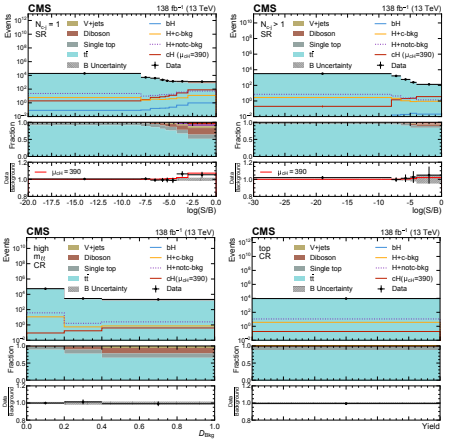


PRL 136 (2026) 011801: event yields vs. $\log_{10}(S/B)$



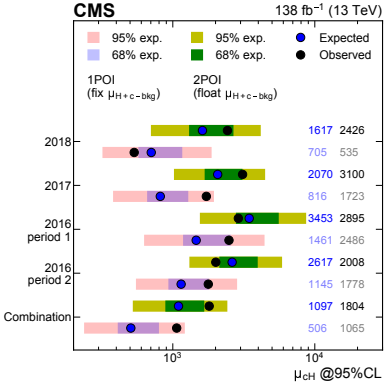
PRL 131 (2023) 041801: boosted $\mathrm{H \to c\bar{c}}$-jet mass, passing region

cH(H→WW): post-fit distributions and per-period limits



1-POI post-fit distributions of the BDT discriminants

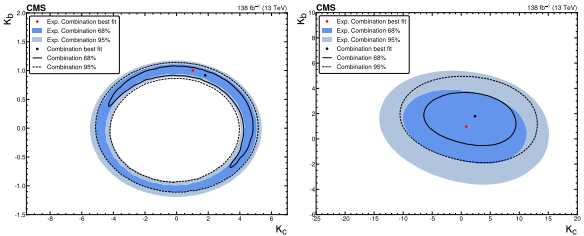
JHEP 08 (2026) 051



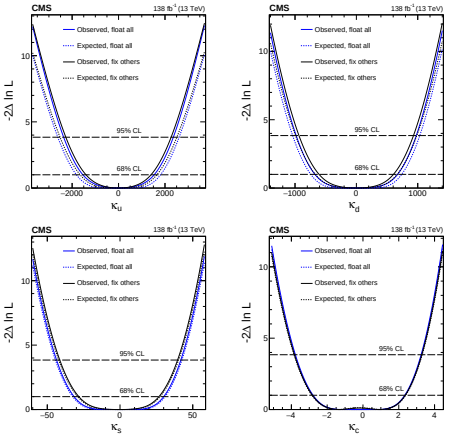
μ_{cH} limits per data-taking period, 1 and 2 POI

Best fit (1 POI): $\mu_{\mathrm{cH}} = 390^{+380}_{-270}$, compatible with SM within 1.5σ; limited by data statistics, then flavour-scheme and ggH heavy-flavour modelling.

Indirect: Higgs p_{T} spectrum and γH



JHEP 04 (2026) 093: (κ_{b}, κ_{c}) from the Run 2 $p_{\mathrm{T}}^{\mathrm{H}}$ combination ($\gamma\gamma$, ZZ, WW, $\tau\tau$); left $\mathcal{B}$ coupling-dependent, right $\mathcal{B}$ unconstrained. Shape sensitive to the sign of κ_{c}.

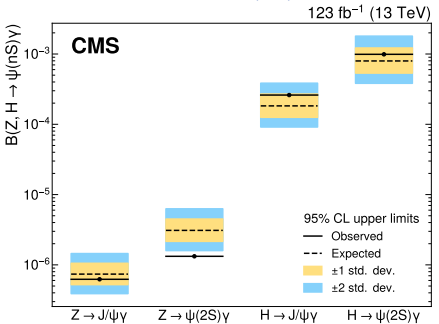


PRD 112 (2025) 112001: light-quark κ scans

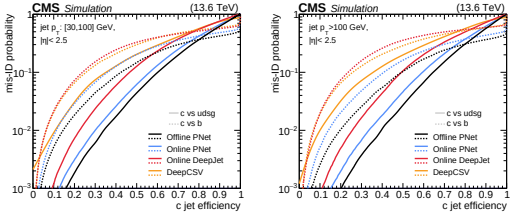
Rare exclusive decays: $H \to J/\psi\gamma$, $\psi(2S)\gamma$

- Direct ($\propto \kappa_c$) and indirect ($H \to \gamma\gamma^* \to \gamma\, J/\psi$) amplitudes interfere $\Rightarrow$ sign sensitivity
- $123\,\mathrm{fb}^{-1}$: $\mathcal{B}(H \to J/\psi\gamma) < 2.6 \times 10^{-4}$, $\mathcal{B}(H \to \psi(2S)\gamma) < 9.9 \times 10^{-4}$
- $\kappa_c/\kappa_\gamma \in (-157, 199)$; with $\kappa_\gamma = 1$: $\kappa_c \in (-166, 208)$

PLB 865 (2025) 139462, CMS-SMP-22-012

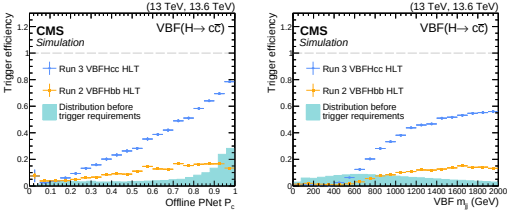


Run 3 HLT flavour tagging



HLT tagger misidentification vs. efficiency

arXiv:2608.18331, CMS-BTV-25-002



VBF H→cc̄ trigger efficiency

References

- VH($\mathrm{H} \to \mathrm{c\bar{c}}$): CMS, PRL 131 (2023) 061801, arXiv:2205.05550 (HIG-21-008)
- $\mathrm{t\bar{t}H}(\mathrm{H} \to \mathrm{c\bar{c}})$ + VH combination: CMS, PRL 136 (2026) 011801, arXiv:2509.22535 (HIG-24-018)
- Boosted $\mathrm{H} \to \mathrm{c\bar{c}}$: CMS, PRL 131 (2023) 041801, arXiv:2211.14181 (HIG-21-012)
- cH(H$\to \gamma\gamma$): CMS, JHEP 11 (2025) 060, arXiv:2503.08797 (HIG-23-010)
- cH(H$\to$WW) and combination: CMS, JHEP 08 (2026) 051, arXiv:2508.14988 (HIG-24-009)
- Differential combination Run 2 (backup): CMS, JHEP 04 (2026) 093, arXiv:2504.13081 (HIG-23-013)
- γH + H$\to 4\ell$ light Yukawas: CMS, PRD 112 (2025) 112001, arXiv:2502.05665 (HIG-23-011)
- Off-shell + on-shell H$\to 4\ell$: CMS, arXiv:2607.23352, submitted to PRD (HIG-25-003)
- H$\to$J/$\psi\gamma$: CMS, PLB 865 (2025) 139462, arXiv:2411.15000 (SMP-22-012)
- HLT flavour tagging: CMS, arXiv:2608.18331 (BTV-25-002)
- $\mathrm{t\bar{t}}$+b/c cross sections: CMS-PAS-TOP-26-003
- HL-LHC: ATLAS and CMS, arXiv:2504.00672 (HIG-25-002)
- BSM: arXiv:2410.05236